

KKIA Terminal Expansion Project

EGM 362 Engineering Project Management

Presented by: Omar Shraim, Hikmat Tawil, Grigor, Yousef Qariab, Kareem Mahmoud

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1- Introduction

Project Definition & Vision 2030

Objective

Expansion of Riyadh airport King Khalid International Airport (KKIA) to increase capacity by **5 million passengers** annually by 2030, by adding a new “smart” terminal.

850

SAR MILLION BUDGET

LEED

PLATINUM TARGET



Major Project Constraints

chedule Time

Alignment with Saudi Vision 2030.

Critical phases must meet 2030 passenger capacity targets.

payments Cost

Strict SAR 850M limit.

Controlling high costs of LEED Platinum and IoT integration.

verified Quality

Compliance with ICAO and Saudi Aviation Laws.

Adherence to LEED sustainability standards.

engineering Resources

Availability of skilled labor for AI and IoT.

Specialized airport-specific infrastructure expertise.

2- Project Plan

Work Breakdown Structure (WBS)

The following are the main deliverables (Level 1), followed by their respective Work Packages (WP):

Project Management & Design

Environmental impact assesment, Regularity approvals and Enginering Design

Site Preparation & Infrastructure

Site Prep & Infrastructure: Land stability, earthworks, and temporary facilities.

Construction

Construction & Services: MEP, Civil works, and Smart Airport device installation.

Urban Planning & Landside Development

Urban Planning: Zoning, parking structures, and environmental impact assessments.

Services

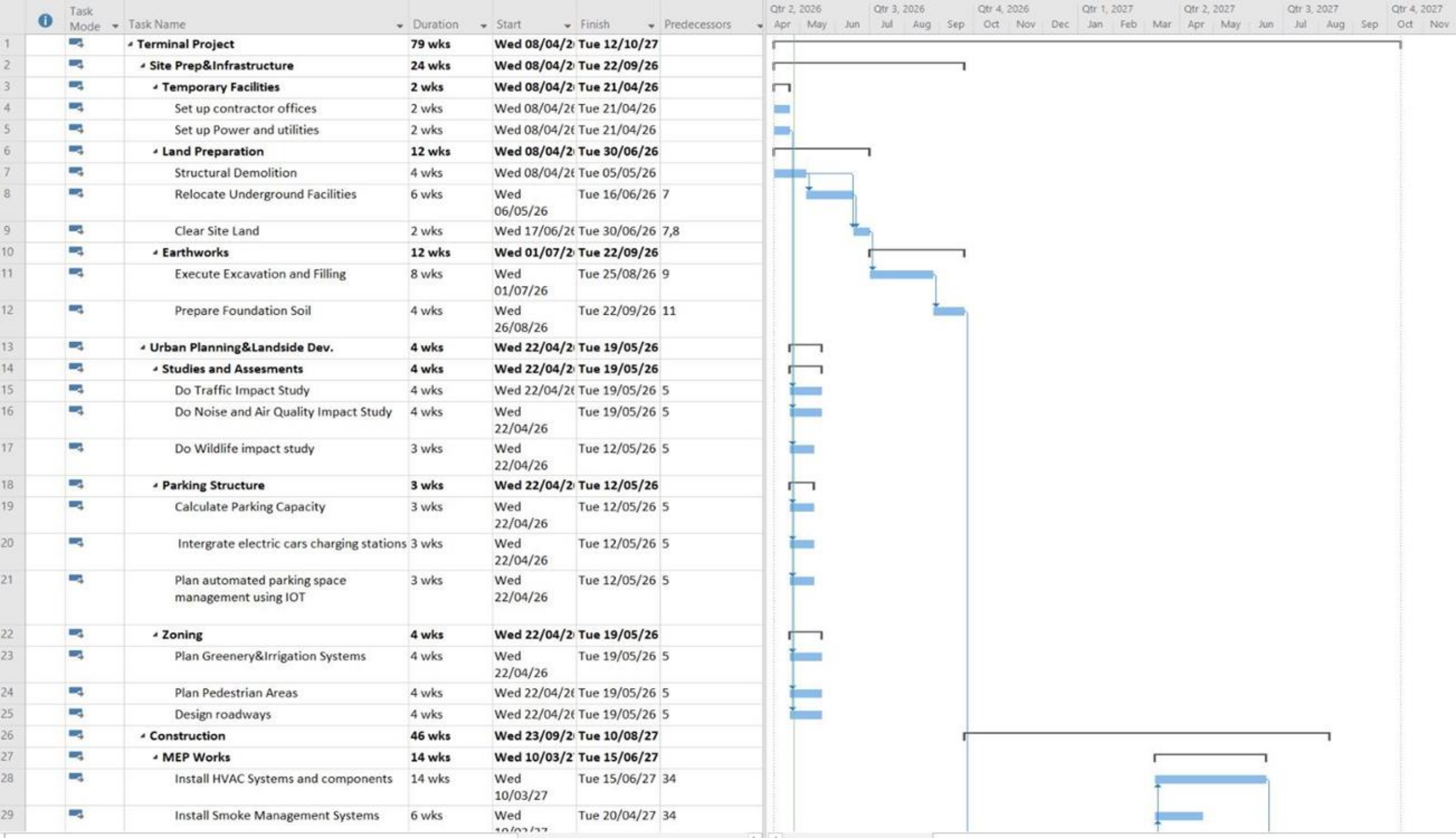
Logistics, Technology, Luxury and Secuirty related Services

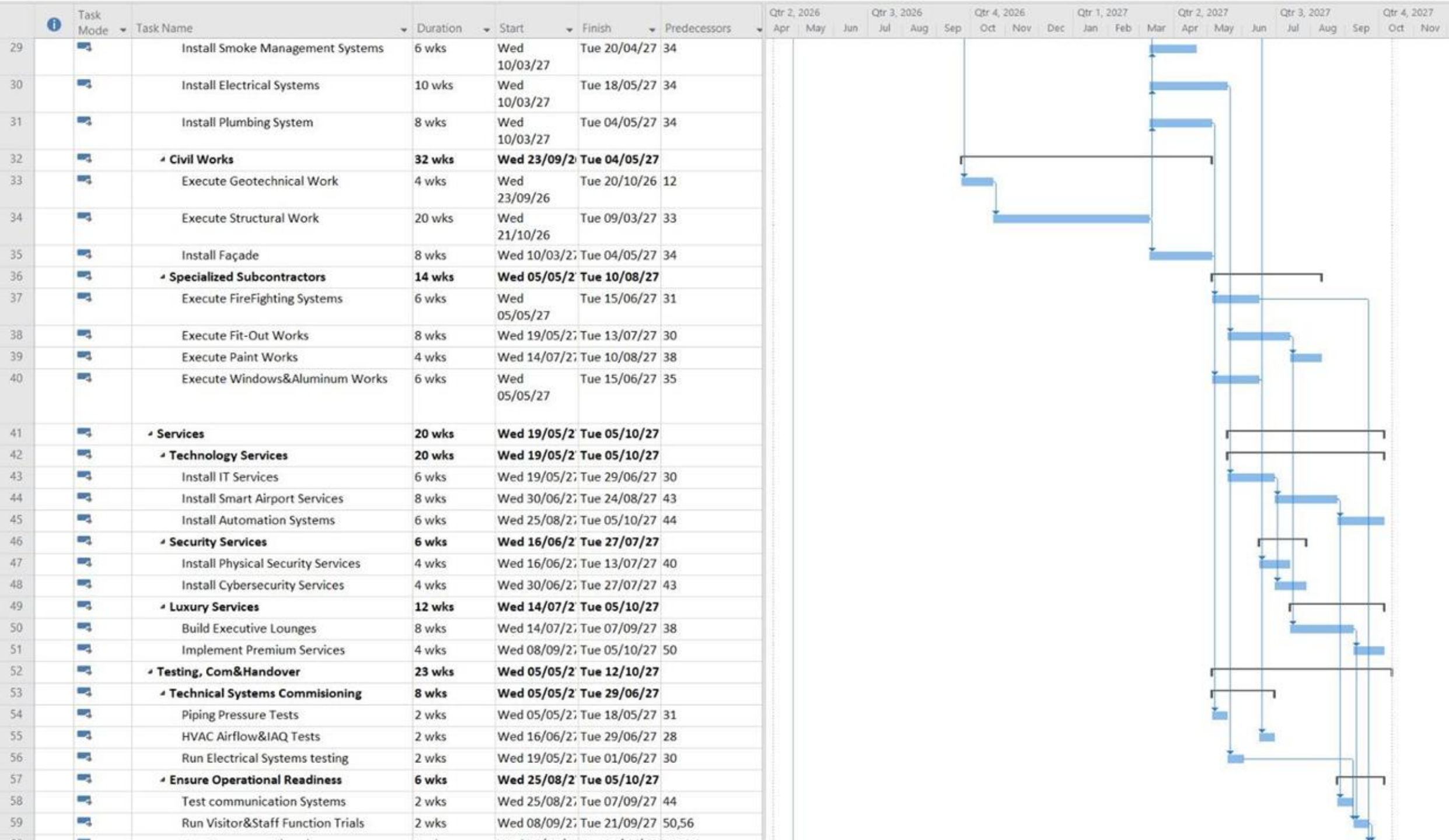
Testing, Commissioning & Handover

Operational Readiness & Airport Transfer (ORAT) , Staff Training & Documentation , Technical Systems Commissioning

Selected Activities Analysis

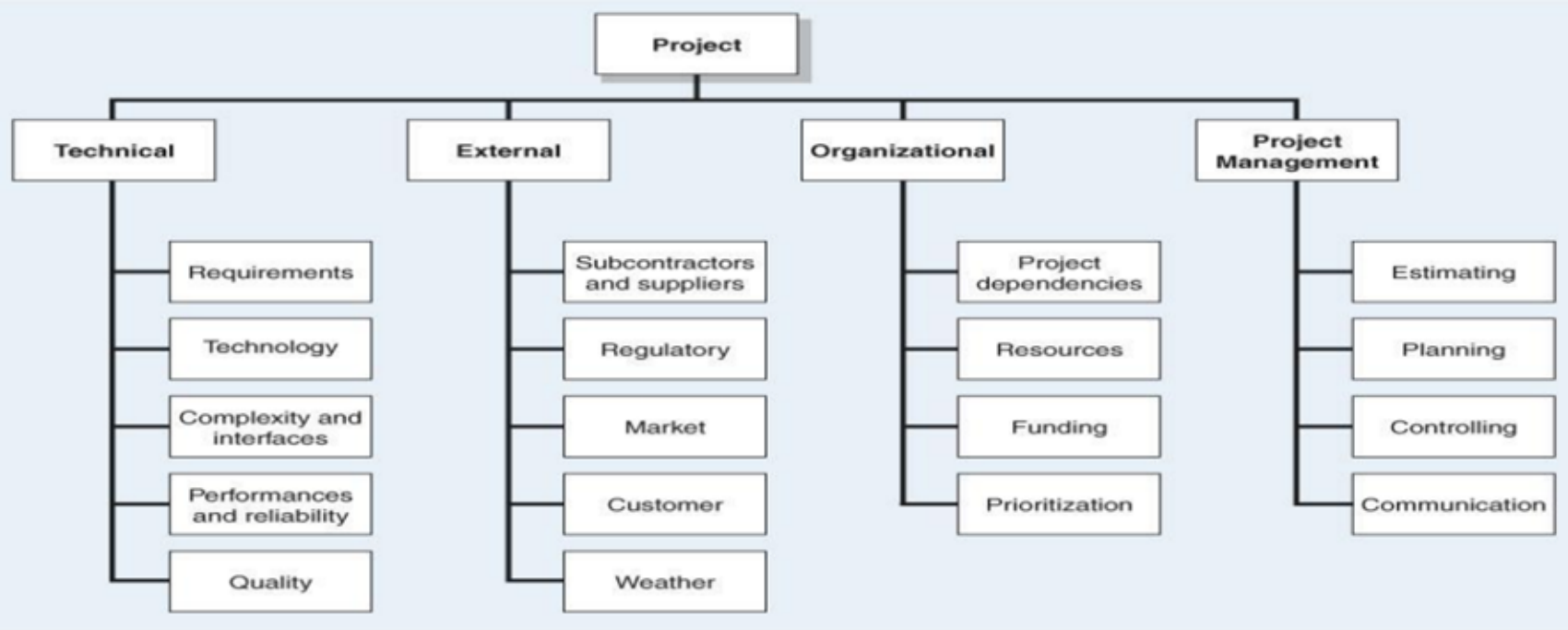
Activity Name	Duration	Estimated Cost	Complexity Drivers
Relocate Underground Facilities	5–6 Weeks	SAR 6 Million	Unknown ground conditions, utility sync
Noise & Air Quality Impact Study	3–4 Weeks	SAR 1 Million	Compliance monitoring, advanced sensors
Install HVAC Systems	12–14 Weeks	SAR 80 Million	High precision, summer heat resilience
Install Smart Airport Devices	6–8 Weeks	SAR 40 Million	AI/IoT integration, Cybersecurity sync





3 – Risk Management Process

1. Risk Identification (RBS)



Risk Categories & Relevant Examples

Technical: Integration of AI/IoT with legacy airport systems.

External: Extreme summer weather and market material inflation.

Organizational: Funding prioritization and stakeholder dependency.

Management: Planning errors and communication gaps.

2. Risk Assessment Matrix (RAM)

Risk Assessment Matrix: A table used to list possible risks and describe them according to the following factors and other factors. Below is an example showing a sample of risks relevant to our project:

Risk	Impact	Chance (L,M,H)	Severity (L,M,H)	Action Trigger	Response Plan
Delayed Approvals	Time and budget overrun	M	L	Report from Civil Authority	Rework and resubmitting approval request
Labor Shortage	Time delay and poor quality	M	H	Numerous and persistent unfilled positions	Expedite search for suitable labor in the local and regional market
Extreme weather (sandstorms/heatwaves)	Delays and material damages	H	H	Weather forecast	Arrange schedule to account for delays (crash schedule if needed)
Material Inflation	Increased project cost	L	M	Financial market reports/news	Search for cheaper resource alternatives/use contingency funds

3. Risk Response Development



Weather Risks

Response: Accept/Retain. Use **contingency funds** for heat-related stoppages; crash schedule if needed.



Material Inflation

Response: Transfer. Sign **fixed-price contracts** with suppliers to "freeze" rates early.



System Failure

Response: Mitigate. Use proven technologies and **early testing** phases to ensure sync.

4- Project Controls

Project Control Methods

Earned Value Management (EVM)

$$SPI = \frac{EV}{PV}$$

Schedule Index: >1 is Ahead of Schedule

$$CPI = \frac{EV}{AC}$$

Cost Index: >1 is Under Budget

Managerial Actions

- ✓ If $SPI < 1$: Crash critical activities to recover time.
- ✓ If $CPI < 1$: Analyze resource efficiency and material waste.
- ✓ **Change Management:** Formal system to prevent scope creep.
- ✓ **Regulatory Authority:** Weekly code approvals with Saudi Civil Defense

5 – Close-out tasks

Project Close out Phases

1. Technical Handover (ORAT)

Resolving Snag Lists, completing MEP commissioning, and securing **Saudi Civil Defense certification**.

In aviation industry: "Operational Readiness and Airport Transfer".

2. Contractual & Financial

Final account settlements, releasing retained bids, and securing the **Taking-Over-Certificate (TOC)**.

3. Administrative Closure

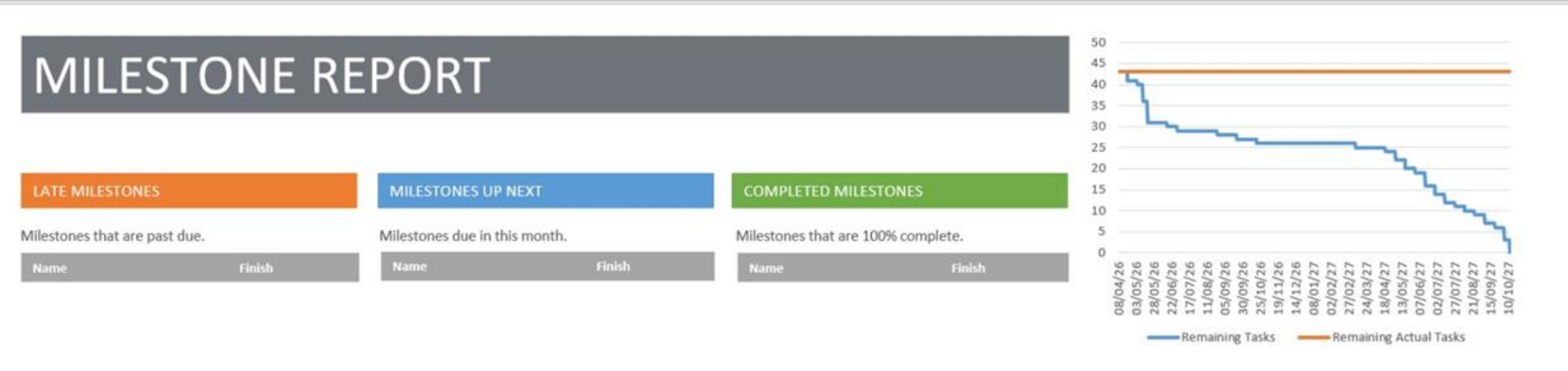
Archiving project charters, handing over user manuals, and formally documenting **Lessons Learned**.

4. Human Resources Release

Demobilizing the site and reassigning engineering staff to mega-projects like **NEOM or the Red Sea Project**.

6 – MS Project

MS Project: Critical Path



Gantt Chart Overview

The project lifecycle spans from **April 2026 to October 2027**. Critical activities identified include:

- ☒ Structural Demolition
- ☒ Relocating Utilities
- ☒ Structural MEP Works
- ☒ Emergency System Testing

Name	Start	Finish	Duration	% Complete	Resource Names
Set up contractor offices	Wed 08/04/26	Tue 21/04/26	2 wks	0%	
Set up Power and utilities	Wed 08/04/26	Tue 21/04/26	2 wks	0%	
Structural Demolition	Wed 08/04/26	Tue 05/05/26	4 wks	0%	
Do Traffic Impact Study	Wed 22/04/26	Tue 19/05/26	4 wks	0%	
Do Noise and Air Quality Impact Study	Wed 22/04/26	Tue 19/05/26	4 wks	0%	
Do Wildlife impact study	Wed 22/04/26	Tue 12/05/26	3 wks	0%	
Calculate Parking Capacity	Wed 22/04/26	Tue 12/05/26	3 wks	0%	
Intergrate electric cars charging stations	Wed 22/04/26	Tue 12/05/26	3 wks	0%	
Plan automated parking space management using IOT	Wed 22/04/26	Tue 12/05/26	3 wks	0%	
Plan Greenery&Irrigation Systems	Wed 22/04/26	Tue 19/05/26	4 wks	0%	
Plan Pedestrian Areas	Wed 22/04/26	Tue 19/05/26	4 wks	0%	
Design roadways	Wed 22/04/26	Tue 19/05/26	4 wks	0%	

7-Conclusion

Conclusions & Lessons Learned



Technical Knowledge

Mastered detailed WBS creation
and accurate resource
estimation for mega-projects.



Regional Insight

Understood the impact of Saudi
Vision 2030 and local
environmental regulations.

Thank You! Any Questions?

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